

ARTICLE

India laying a foundation for the next generation

COMMENT



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WHAT is common to Microsoft, IBM and Alphabet (the parent company of Google)? The answer is that the chief executives are people of Indian origin who have graduated from India's higher education system.

From ancient times, Indian civilisation and culture has placed a social premium on education. The concept of zero was developed by the Indian mathematician Aryabhata who lived in the fifth century. The word "guru" comes from Sanskrit and literally means the dispeller of darkness.

India has always valued its teachers and in the "gurukul" system of education, the relation between the teacher and their learners was held as sacred.

As India steps into the 75th year of its independence and we count the joys and sorrows of this momentous journey, we should also reflect on the legacy this generation will leave for future generations.

Like other developing countries and emerging economies, India faces daunting challenges in delivering health, education and a life of equal and fair opportunity to its vast and diverse population. But economic growth and social progress is possible only with human development.

India has made significant progress in education in the past 75 years. India's technological institutions, notably the Indian Institutes of Technology (IITs) compare with the best in the world in terms of the quality of their students. There has been a huge



INDIAN Space Research Organisation robot 'Vyomamitra' on display at an International conference on Human Space flight and Exploration and The Challenges and Future Trends in Bangalore, India, in January 2020. The robot was designed to simulate human functions before real astronauts go to space. Despite many developmental challenges, India has invested in hi-technology, says the writer. I EPA

expansion at all levels of the education system in India. For example, in 2015 there were 799 universities in India. By last year this figure rose to 1 043.

The expansion of the public education system, with low fees and an augmented programme of affirmative action, has enabled greater inclusiveness in the student body that is addressing gender, social and regional inequalities and imbalances.

Science and education have powered India's growth story. It was in this context that India offered 50 000 scholarships to African students over five years during the last India Africa Forum Summit.

In the 1960s, India was on the verge of a mass food shortage and the country was dependent on foreign food aid. The government launched the "Green Revolution" with Indian agricultural scientists like MS Swaminathan who joined hands with Norman Borlaug and others to usher in high-yielding varieties of rice and wheat.

In the decades since, India's agriculture has modernised and, last year ,agricultural exports from India soared

to a record \$19 billion. The next stop of the government was the dairy sector. The engineer Verghese Kurien spear-headed a dairy co-operative movement that came to be known as the "White Revolution". It led India from being a milk-deficient country to becoming the world's largest dairy producer. In the process, it has given remunerative prices to millions of dairy farmers across the country while nourishing its large population.

Despite multiple developmental challenges, India has invested precious resources in hi-technology. The Indian Space Research Organisation (ISRO) successfully launched an inter-planetary mission, Mangalyaan or the Mars Orbiter Mission, becoming the first country to do so on its first attempt.

India's lunar probe Chandrayaan 1 was launched successfully at a fraction of the cost of similar programmes of other countries.

South Africa has provided telemetry, tracking and control services for India's lunar and Mars missions. ISRO is one of six government space agencies in the world which possess

full-launch capabilities, deploy cryogenic engines, launch extraterrestrial missions and operate large fleets of artificial satellites. India's advanced atomic energy programmes is based on a three-stage programmes that seeks to secure the country's long-term energy independence. It has an impeccable safety and security record.

Nowhere does science and development intersect more materially for people than in the health sector. India's booming pharmaceutical sector is a recognised innovator and producer of life-saving medicines at affordable cost. India is the world's largest supplier of low-cost generics and is among the largest producers of drugs, both in value and volume terms.

It has the second largest number of US Food and Drug Administration-approved manufacturing plants.

The IT sector in India grew in the 1990s and demonstrated how a developing country could leapfrog to achieve a position of strength in an advanced technology industry. India's IT industry accounts for approximately 55% of the global service sourcing market (\$200bn to \$250bn) in 2019/20.

India is home to the world's third largest technology start-up ecosystem, with 11 000 to 12 500 start-ups incepted in 2015 to 2020. It also has the third largest number of unicorns in the world as of 2020.

Against all odds, India's democracy, economy and society have prospered over the 75 years. We have benefited from our engagement with the world, including with South Africa, with whom we share the values of diversity, democracy and development.

Today, a new generation in India is on the ascendant that will deal with the country's challenges and opportunities and take the country to new frontiers. It possesses the skills and resources to do so successfully.

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DESCRIPTION

India Laying a Foundation for the Next Generation:

India has always placed great importance on education, from the ancient gurukul system to the modern world-class institutions that produce global leaders. Indian-origin CEOs now lead major companies like Microsoft, IBM and Alphabet, showing the strength of India's education system. In the 75 years since independence, India has made remarkable progress by expanding universities, improving inclusiveness and encouraging scientific and technological growth. The country overcame early challenges such as food shortages through the Green Revolution

and later transformed the dairy sector through the White Revolution. India also invested in high technology despite limited resources, achieving major successes like the Mars Orbiter Mission and Chandrayaan at very low cost. The nation is now a leader in affordable medicines, with a strong pharmaceutical industry, and its IT sector and start-up ecosystem are among the world's largest. Today, a confident and skilled new generation is rising, equipped with education, innovation and opportunities ,ready to take India to new heights and shape a brighter future for the nation.

